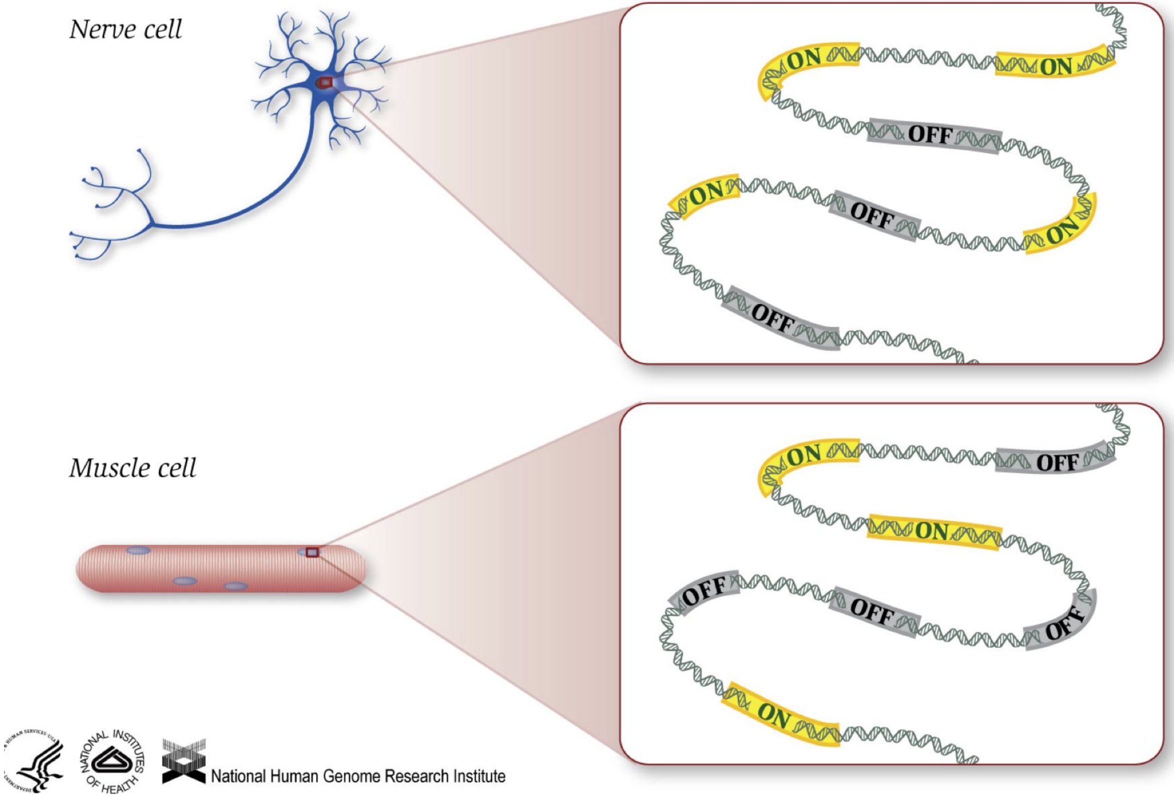
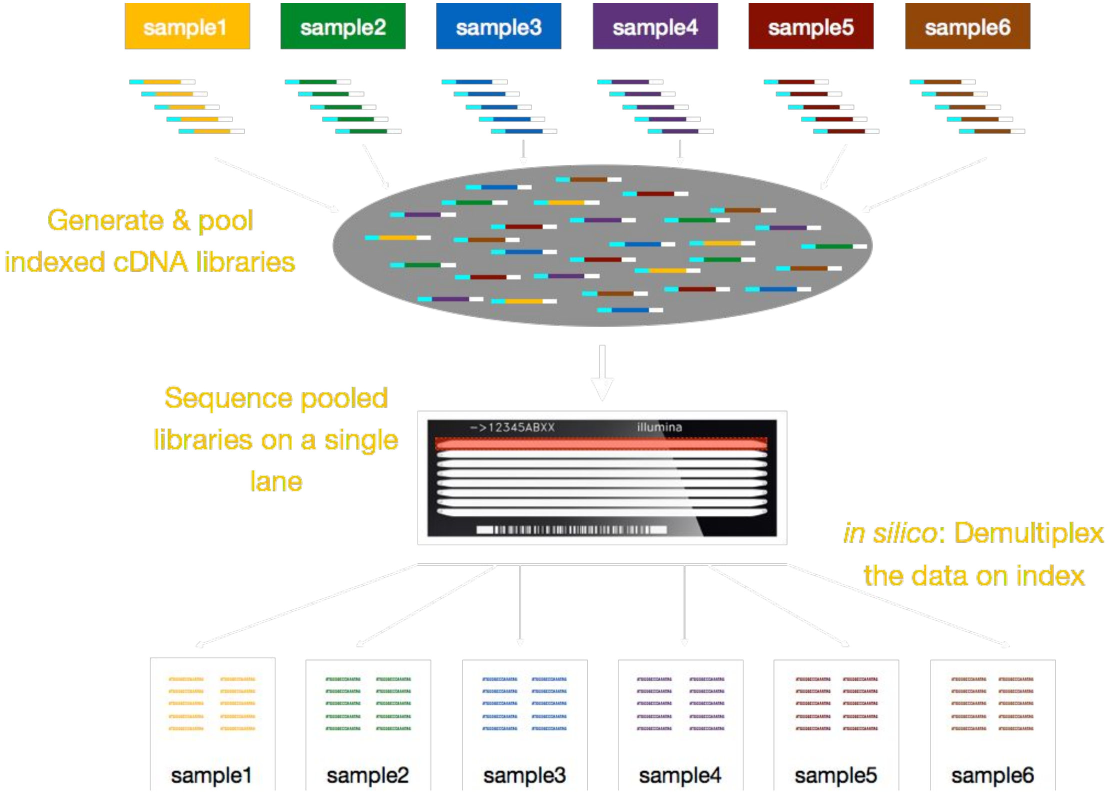
Day 2: RNA-Seq Workflow

Adapted from: https://hbctraining.github.io/Intro-to-bulk-RNAseq/schedule/links-to-lessons.html

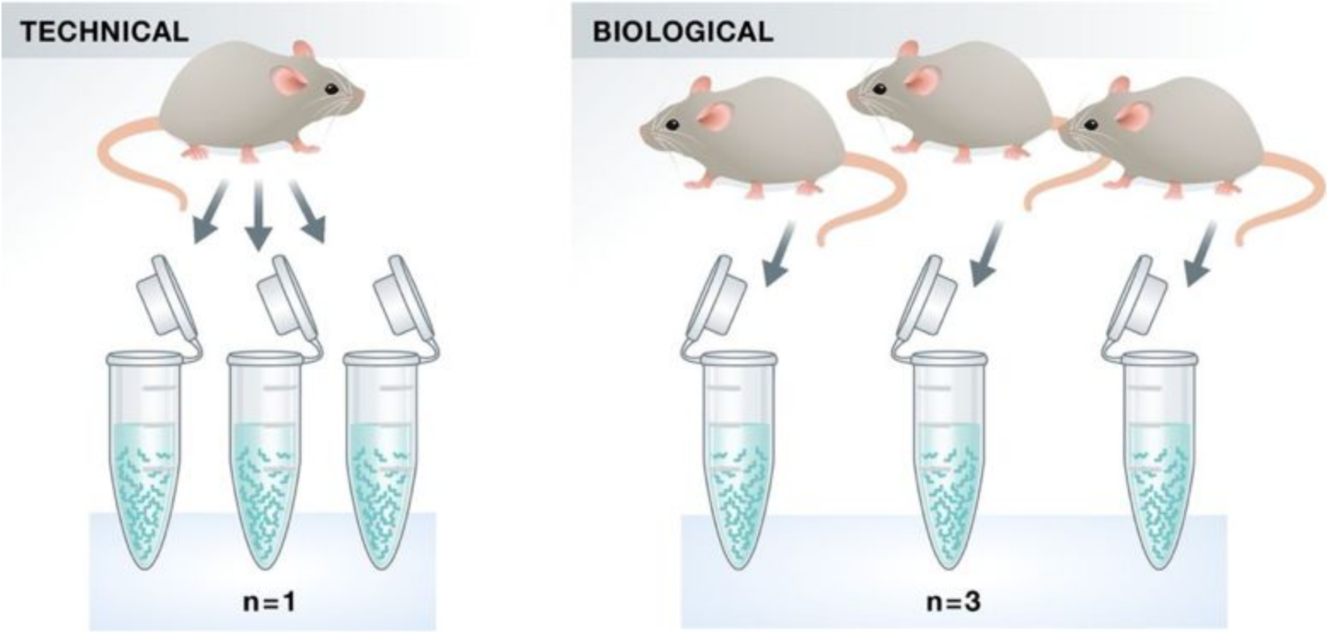
Why RNA-seq?


Nerve cell
ON
ON
OFF
ON
OFF
ON
OFF
Muscle cell
ON
OFF
ON
OFF
OFF
OFF
ON
National Human Genome Research Institute

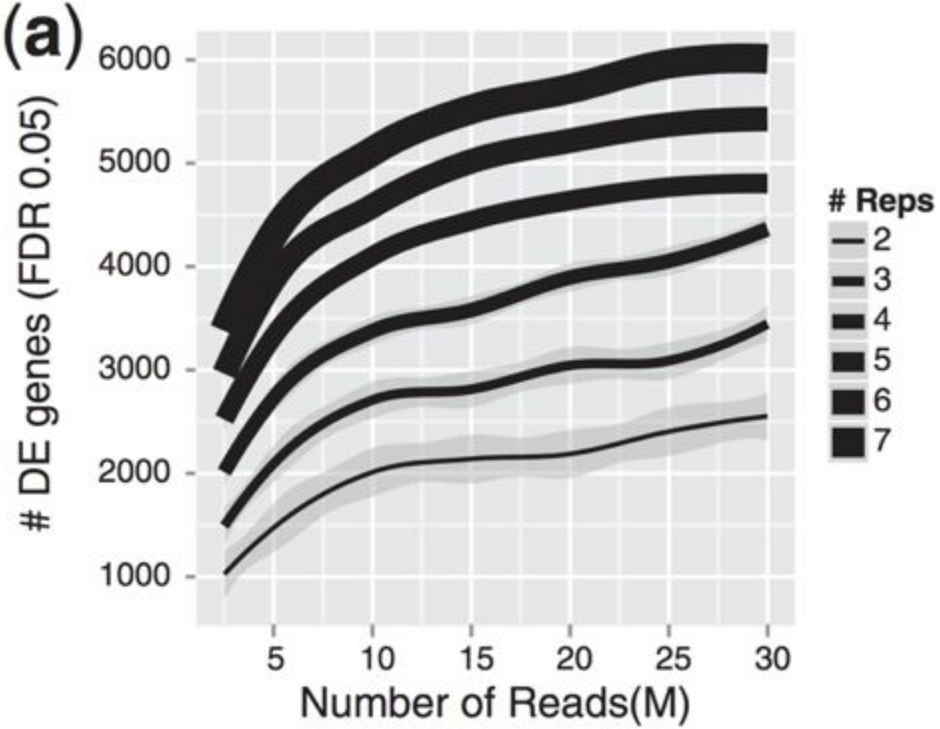
Sequencing your samples



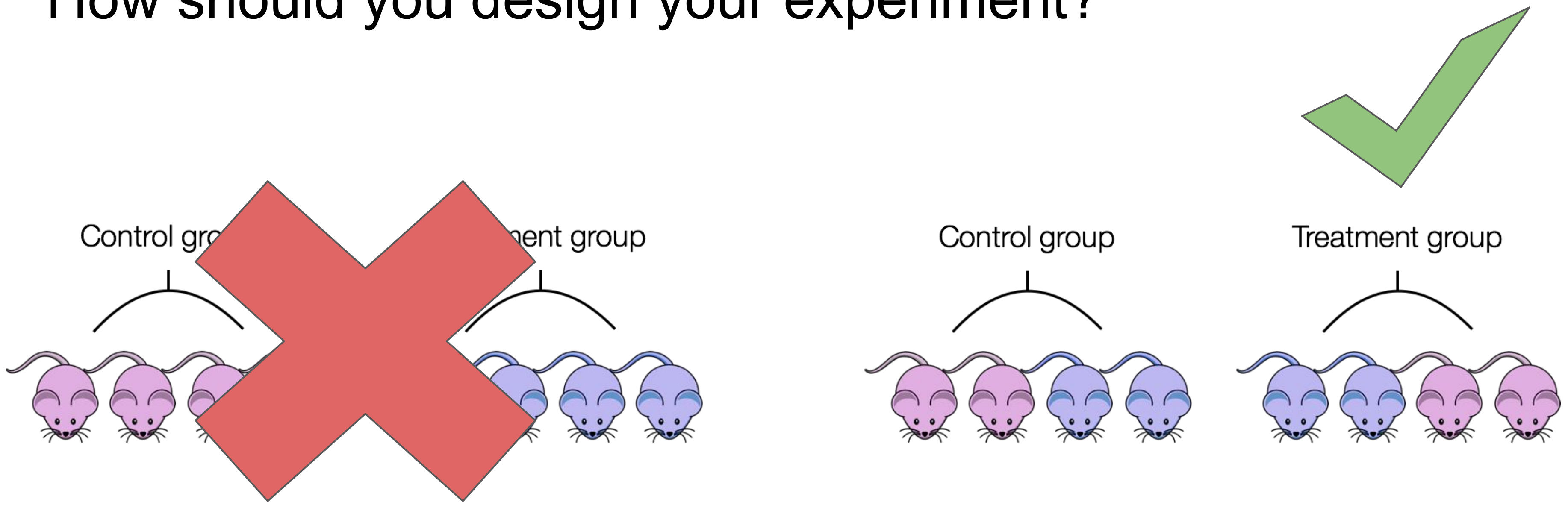
About Replicates



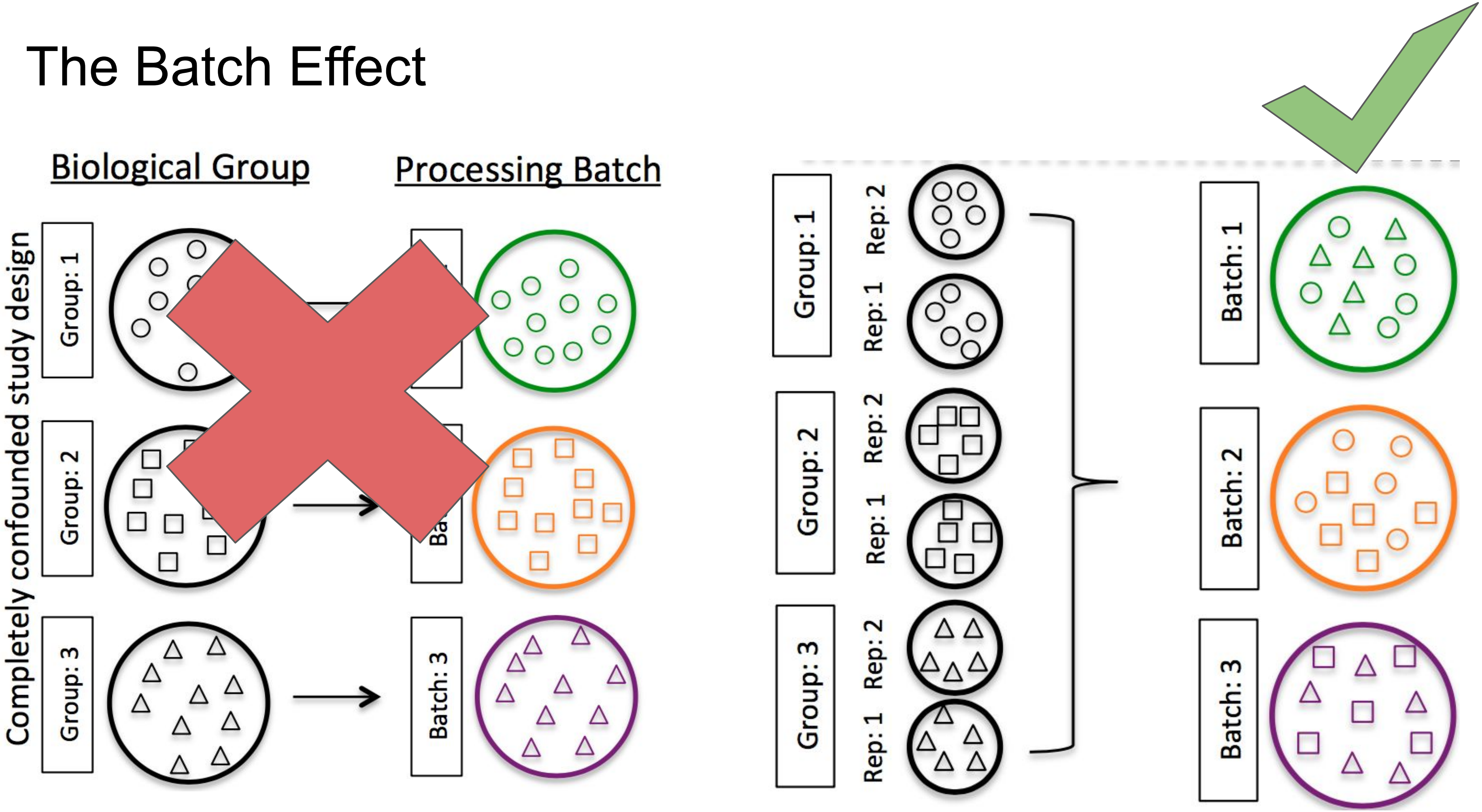
Why Replicates and sequencing depth matters



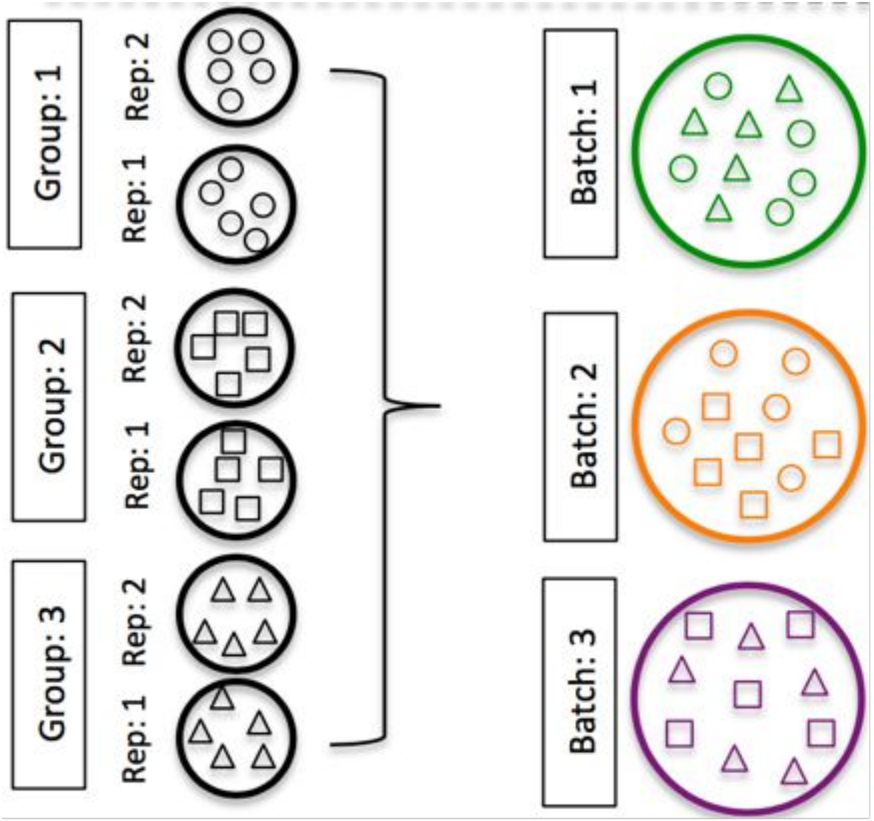
How should you design your experiment?

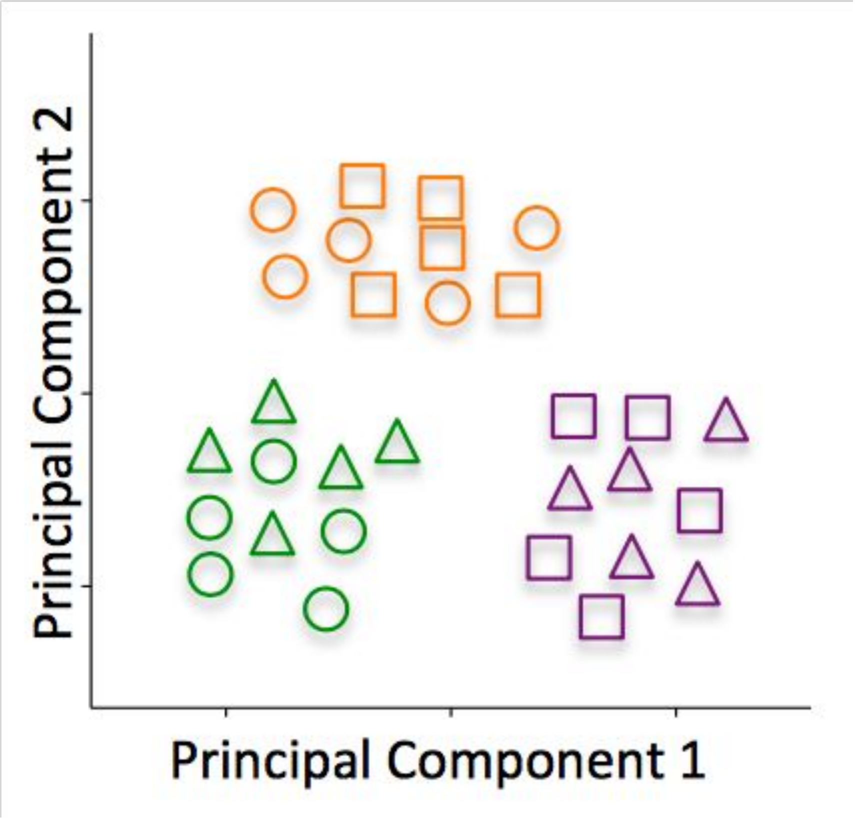


The Batch Effect

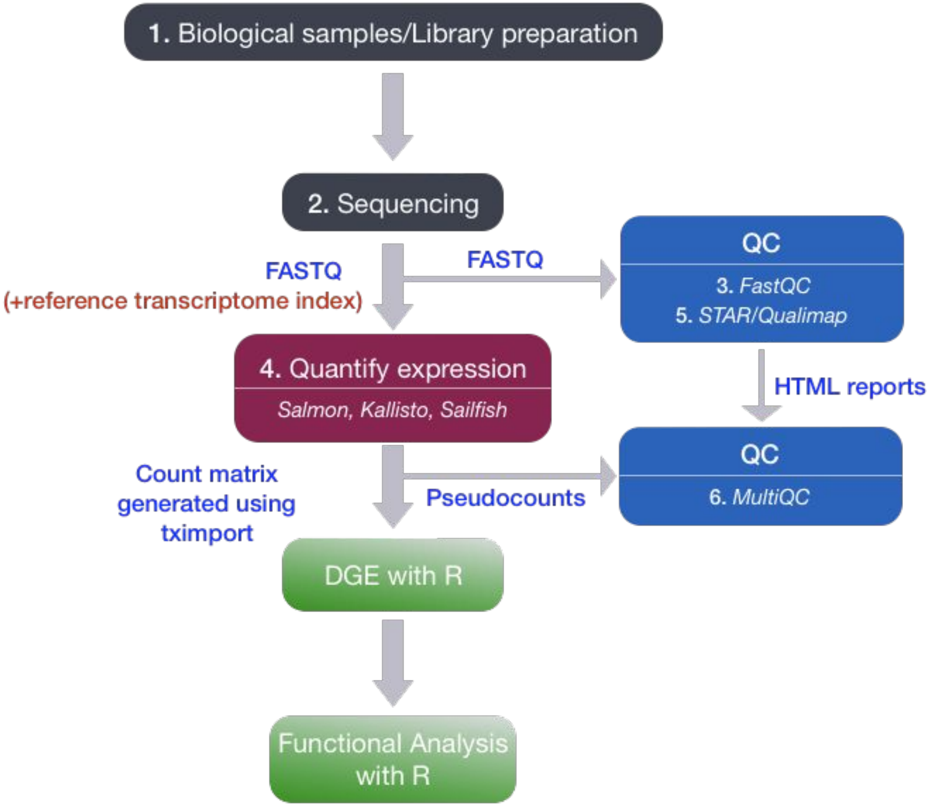


Reducing the Batch Effect





Gene quantification



What's in a FASTQC file?

```
@HWI-ST330:304:H045HADXX:1:1101:1111:61397

CACTTGTAAGGGCAGGCCCCCTTCACCCTCCCGCTCCTGGGGGANNNNNNNNNNANNNCGAGGCCCTGGGGTAGAGGGNNNNNNNNNNNNNNGATCTTGG

+

@?@DDDDDDHHH?GH:?FCBGGB@C?DBEGIIIIAEF;FCGGI#########################################################
```

Line	Description
1	Always begins with '@', followed by information about the read
2	The actual DNA sequence
3	Always begins with a '+', and sometimes the same info as in line 1
4	Has a string of characters representing the quality scores; must have same number of characters as line 2

Measuring quality

```
Quality encoding: !"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHI
                  |         |         |         |         |
   Quality score: 0.........10........20........30........40
```

Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10,000	99.99%

Measuring quality with fastqc

```
fastqc -t 5 ${FASTQ_FILE}.fastq.gz -o STATS_DIR/fastqc_raw
```

```
(base) helenaconceicao@OSXLAP13259 fastqc_raw % tree .
.
├── RNAseq_NCC_D0_WT_1_R1_001_fastqc.html
├── RNAseq_NCC_D0_WT_1_R1_001_fastqc.zip
├── RNAseq_NCC_D0_WT_1_R2_001_fastqc.html
├── RNAseq_NCC_D0_WT_1_R2_001_fastqc.zip
├── RNAseq_NCC_D0_WT_2_R1_001_fastqc.html
├── RNAseq_NCC_D0_WT_2_R1_001_fastqc.zip
├── RNAseq_NCC_D0_WT_2_R2_001_fastqc.html
├── RNAseq_NCC_D0_WT_2_R2_001_fastqc.zip
├── RNAseq_NCC_D0_WT_3_R1_001_fastqc.html
├── RNAseq_NCC_D0_WT_3_R1_001_fastqc.zip
├── RNAseq_NCC_D0_WT_3_R2_001_fastqc.html
├── RNAseq_NCC_D0_WT_3_R2_001_fastqc.zip
```

Investigating the fastqc output

Summary

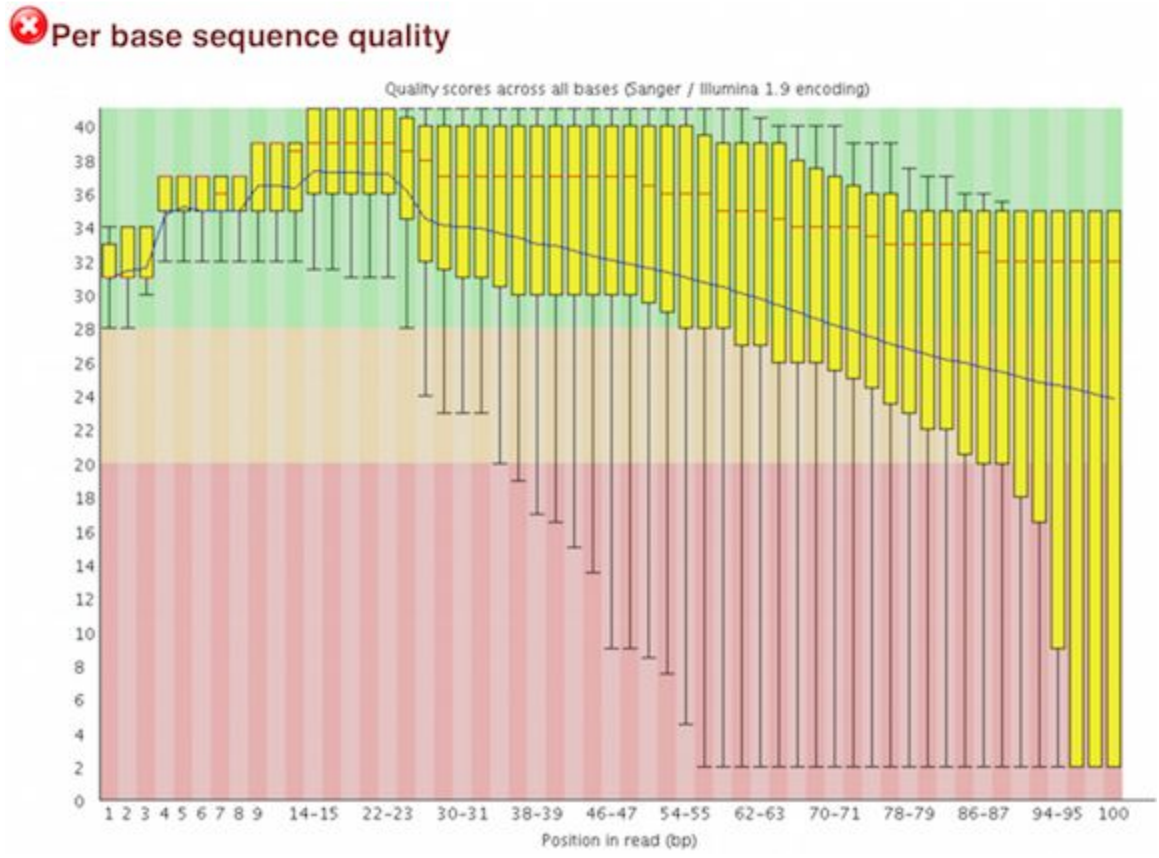
- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

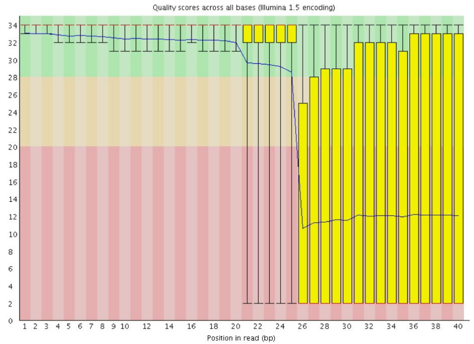
Basic Statistics

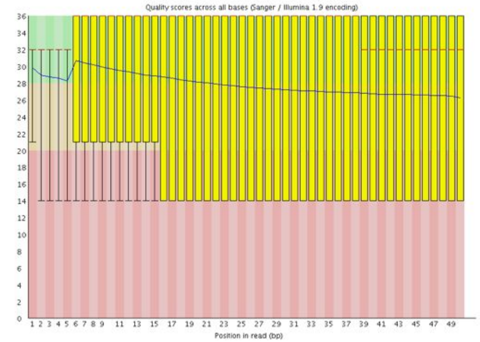
Measure	Value
Filename	RNAseq_NCC_D0_WT_1_R1_001.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	21191722
Sequences flagged as poor quality	0
Sequence length	101
%GC	48

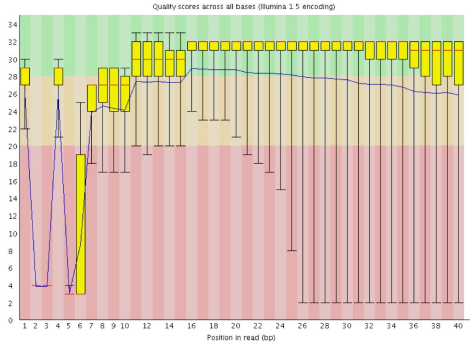
>20 million reads in a glance

Some bad examples

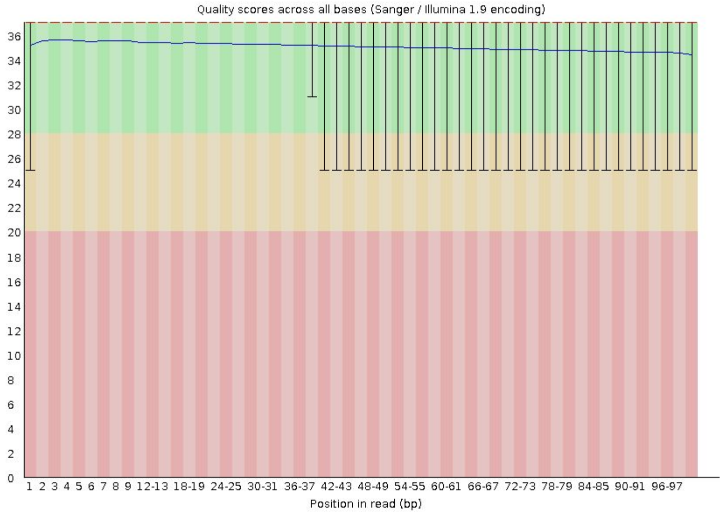








Some good examples



Trimming the adaptors

```
cutadapt \
    -a AGATCGGAAGAGCACACGTCTGAACTCCAGTCAC \
    --minimum-length=25 -j 5 \
    -o trimmed_R1.fastq.gz -p trimmed_R2.fastq.gz \
    R1.fastq.gz R2.fastq.gz > report_file.txt
```

this is optional for RNAseq, since the aligners made for RNAseq are able to align reads partially, to be able to account for splicing

Aligning the reads against the genome

```
hisat2 \
    -p 5 \                                  #number of threads
    --phred33 \                             #quality filter
    --rna-strandness RF \
    --dta \                                 #optional
    --no-unal \                             #optional
    -x $GENOME_INDEX_DIR \
    -1 trimmed_R1.fastq.gz -2 trimmed_R1.fastq.gz \
    --summary-file hisatSummary.txt |
        samtools view -bS -@ 5 - |
        samtools sort -@ 5 -o sample_name.bam -
```

Hisat summary output

```
21753262 reads; of these:
  21753262 (100.00%) were paired; of these:
    3752759 (17.25%) aligned concordantly 0 times
    16817160 (77.31%) aligned concordantly exactly 1 time
    1183343 (5.44%) aligned concordantly >1 times
    ----
    3752759 pairs aligned concordantly 0 times; of these:
      424783 (11.32%) aligned discordantly 1 time
    ----
    3327976 pairs aligned 0 times concordantly or discordantly; of these:
      6655952 mates make up the pairs; of these:
        3651889 (54.87%) aligned 0 times
        2499841 (37.56%) aligned exactly 1 time
        504222 (7.58%) aligned >1 times
91.61% overall alignment rate
```

Creating count matrix

```
featureCounts \
        -T 5 \
        -t exon \
        -s 2 \
        -g gene_id \
        -a genes.gtf \
        -o featureCounts.txt \
        -p \
        sample_name.bam
```

Now to our activity!

https://simoes-costa-lab.github.io/genomics-course-usp2026/rnaseq/practical.html

Day 3 | 30 Jun 2026

Time	Type	Location	Activity
09:00–10:30	Lecture	Queijinho A5	**Lecture 03:** Gene Regulatory Networks (MSC)
11:00–12:00	Lecture	Queijinho A5	**Lecture 04:** ATAC-seq and CUT&RUN (APA)
13:30–14:15	Science Talk	Multimedia Room (B1)	*Metabolism and Gene Regulation* (Fjodor Merkuri)
14:30–16:30	Bootcamp	Multimedia Room (B1)	RNA-seq Data Analysis (Helena Conceição)
16:30–18:00	Project	Multimedia Room (B1)	Project Development